Wang et al. 2019 — Glioblastoma

Brain · 10x · 9,616 malignant cells (7/10 samples) · embryonic-program cell-state analysis

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/10 samples (≥396 cells) intermix on one continuous manifold (radius-graph: ~34 detached cells removed → 1 component; iLISI 1.0→2.35/7). θ=2 archipelago dissolves by θ=10; the pSom↔Crest axis reproduces within all 7 patients (r=−0.50 to −0.64).

Gate 2 — module geography → GO *(non-branching)*

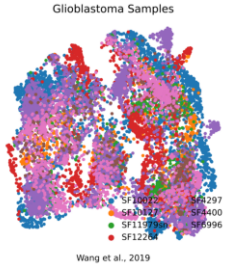
pSom declines along pseudotime (ρ=−0.36) and Crest rises to a terminal readout (ρ=+0.29; Crest-high decile mean pt=0.88); Gast a weak co-gradient at the origin. Non-branching case.

Gate 3 — hub cross-repression → NA

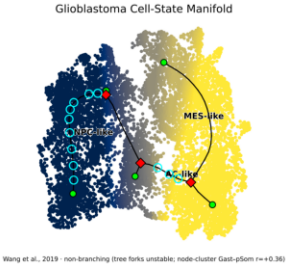
PPT tree fit + tested: forks unstable across N (0–3); node-clusters CO-VARY (Gast↔pSom r=+0.29), hub is co-high not two arms; per-gene r<0 is an XOR-gene artifact (single-module split-half null=+0.5). No cross-repressing branch pair → not tested. Curve, not tree.

STEP FIGURES

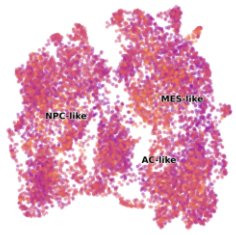
Gate 1: integration



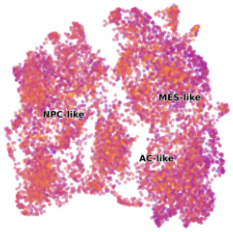
Trajectory / tree



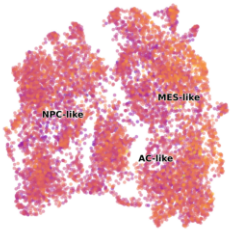
Gast



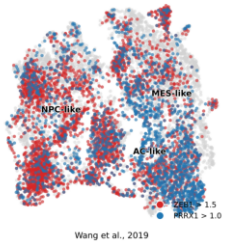
pSom



Crest



EMT-TFs



Dataset. Glioblastoma (Wang et al., 2019), 10x — 10 patients / 16,317 cells (master scope); malignant only: 9,916 loaded, 9,650 across 7 samples (≥396) after the count-gap threshold, ~9,616 after island removal enter the trajectory; whole malignant manifold (no lobe scoping — cycling orthogonal).

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The manifold is multipolar (NPC/AC/MES Neftel states), but a fitted PPT tree gives no genuine bifurcation: forks are node-number unstable and its node-clusters CO-VARY in Gast↔pSom (r=+0.29, both opposing Crest), with a single co-high hub rather than a Gast-arm and a pSom-arm. So the embryonic axis is one continuous pSom → terminal-Crest gradient (NPC/OPC origin → MES/hypoxia neural-crest terminus), rooted at the pSom/Gast pole; cell cycle orthogonal (ρ=+0.07), whole manifold kept. EMT-TFs ZEB1 (root) → PRRX1 (terminus); SNAI1/FOXC below the 3% floor (TWIST1 4.3%, weak). Module means Gast −0.003 / pSom +0.008 / Crest −0.007.

Parameter choices

Min cells/sample	≥396 (keeps 7/10; gap 396→143)	**Integration**	Harmony θ=10; PCA50; n_pcs=20
Scope	whole manifold (no scoping)	**Trajectory**	curve N=6 (tree tested: forks unstable)
Root node	curve tip 2 = pSom/Gast-high, Crest-low	**Cell states**	AC/NPC/MES-like (Neftel 2019)